

LAI MINH SANG

Data Engineer | Software Developer

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PROFESSIONAL SUMMARY

Software Developer with 5+ years of backend experience and hands-on data engineering projects. Strong in Java, Python, SQL, Spark DataFrame, ETL/ELT, RDBMS, and NoSQL. Practiced Airflow/NiFi orchestration, Kafka fundamentals, Docker, Git/CI basics, AWS data services, basic Data Quality checks, metadata/Data Catalog concepts, Delta Lake/Iceberg, Elasticsearch, and distributed data processing.

TECHNICAL SKILLS

Languages: Python, Java, SQL; basic Scala reading ability

Data Processing: Apache Spark DataFrame, PySpark, Parquet, partitioning, caching, batch ETL/ELT

Orchestration / Streaming: Airflow, Apache NiFi, Kafka fundamentals: topics, producers/consumers, consumer groups

Databases: MySQL, PostgreSQL, AWS RDS, Redshift; MongoDB; data warehousing and dimensional modeling

DQ / Governance: Schema validation, null/duplicate checks, referential checks, reconciliation, metadata/Data Catalog basics

Cloud / DevOps: AWS S3, Glue, Redshift, RDS; Docker, Docker Compose, Git/GitHub, CI basics; K8s/HDP awareness

Exposure: Delta Lake/Iceberg fundamentals, Elasticsearch basics, Linux, REST APIs

PROFESSIONAL EXPERIENCE

Freelance Software Engineer / Data Engineering Projects

2025-Present

Remote

- Built pipeline and warehouse practice projects using Python, SQL, Spark, Airflow/NiFi, Docker, MySQL/PostgreSQL, and MongoDB.
- Designed ETL/ELT flows from raw ingestion to curated analytical tables with schema validation, cleansing, aggregation, and reconciliation checks.
- Containerized local data platforms with Docker Compose and documented setup, pipeline execution, troubleshooting, and deliverables in Git/GitHub.

Software Developer, FUNiX

2022 - 2024

Hanoi / Remote

- Developed and maintained backend services and internal web applications, strengthening API design, database modeling, debugging, and deployment support.
- Worked with Java/Spring Boot, SQL databases, REST APIs, authentication flows, and collaborative Git workflows; communicated with stakeholders to clarify requirements and resolve issues.

SELECTED DATA ENGINEERING PROJECTS

E-commerce Data Pipeline

PySpark, AWS Glue, S3, Redshift, RDS, SQL

- Designed an end-to-end ETL pipeline from raw e-commerce data ingestion to structured warehouse tables for analytical querying.
- Developed PySpark DataFrame jobs for cleansing, transformation, aggregation, and loading processed datasets to S3/Redshift/RDS; applied DQ checks including required fields, duplicates, type validation, and row-count reconciliation.

Log Analytics and Anomaly Detection Pipeline

PySpark, Parquet, PostgreSQL, AWS S3

- Parsed simulated Nginx/application logs to extract IP, URL, status code, timestamp, and request metrics from semi-structured data.
- Calculated KPIs including request volume, error rate, status-code distribution, and suspicious traffic patterns; stored curated data in Parquet and served metrics through PostgreSQL.

Large-Scale Student Performance Analysis

Spark DataFrame, SQL, Parquet

- Processed tens of millions of simulated exam-score records using Spark DataFrame operations, cleansing, joins, grouping, and ranking logic; improved batch jobs with partitioning and caching.

University Data Warehouse / ETL Labs

NiFi, MySQL, MongoDB, Dimensional Modeling

- Built lab pipelines for dimension/fact loading, SCD Type 2, transaction facts, periodic snapshots, accumulating snapshots, factless facts, RDBMS permissions, JDBC pools, MongoDB CRUD, and SQL validation.

ACADEMIC BACKGROUND

FUNiX Online Education - Data Engineer Program

2024

- Specialized in Data Engineering principles, Spark, AWS data services, RDBMS/NoSQL, ETL/ELT, and data warehousing fundamentals.

FPT Polytechnic - Bachelor's Degree in Software Engineering / Software Application

2017-2020

- Developed a strong foundation in software engineering, backend development, database design, and practical application development.